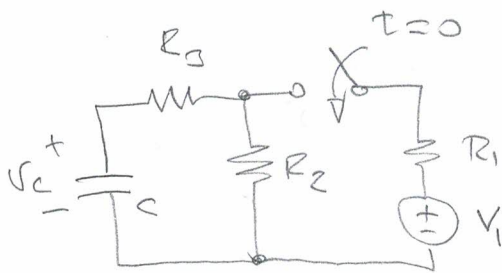


5.37)



$$V_1 = 10V$$

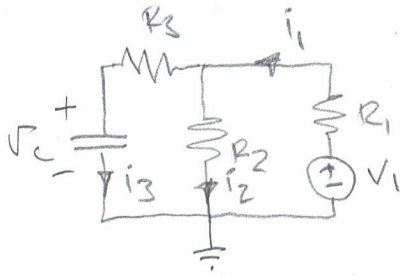
$$C = 200 \mu F$$

$$R_1 = 300 \text{ m}\Omega$$

$$R_2 = R_3 = 1.2 \text{ k}\Omega$$

Find  $V_C(t)$  for  $t > 0$

Lösung:



KSPL ger

$$\begin{cases} -V_C - R_3 i_3 + R_2 i_2 = 0 & (1) \end{cases}$$

$$\begin{cases} -V_1 + R_1 i_1 + R_2 i_2 = 0 & (2) \end{cases}$$

DRSSUNG gitter ATT

$$i_1 = i_2 + i_3 \quad (3)$$

$$\text{oder} \quad V_C = V_C(0) + \frac{1}{C} \int i_3 dt \quad (4)$$

$$(4) \text{ i(1)} \quad -V_C(0) - \frac{1}{C} \int i_3 dt - R_3 i_3 + R_2 i_2 = 0 \quad (5)$$

(2) & (3) ger

$$-V_1 + R_1 (i_2 + i_3) + R_2 i_2 = 0$$

$$R_1 i_2 + R_1 i_3 + R_2 i_2 = V_1$$

$$i_2 = \frac{V_1 - R_1 i_3}{R_1 + R_2} \quad (6)$$

(6) i(5) ger

$$-V_C(0) - \frac{1}{C} \int i_3 dt - R_3 i_3 + \frac{R_2}{R_1 + R_2} (V_1 - R_1 i_3) = 0$$

$$-V_C(0) - \frac{1}{C} \int i_3 dt - R_3 i_3 - \frac{R_2 R_1}{R_1 + R_2} i_3 + \frac{R_2 V_1}{R_1 + R_2} = 0 \quad (6.5)$$

DERIVIERE MAP t

$$\frac{1}{C} i_3 + \left( R_3 + \frac{R_2 R_1}{R_1 + R_2} \right) \frac{di_3}{dt} = 0$$

oder som

$$\frac{di_3}{dt} + \frac{1}{C \left( R_3 + \frac{R_2 R_1}{R_1 + R_2} \right)} i_3 = 0 \quad (7)$$

$= 1/\tau$

Som här lösning

MED SIFFROR

$$i_3 = A e^{-t/\tau}$$

$$\tau = 0.240 \text{ sek}$$

ÅTERSTÄLL KONSERVATION A. UTVÄRDERA (6.5) VID  $t=0$ 

$$\underbrace{-V_c(0)}_{=0} - \underbrace{\frac{1}{C} \int_0^t i_3 dt}_{=0} - \left( R_3 + \frac{R_2 R_1}{R_2 + R_1} \right) \underbrace{i_3}_{=A} + \frac{R_3}{R_1 + R_2} V_1 = 0$$

OLADDAD VID  $t=0$        $\int_0^t \dots dt$

$$\Rightarrow A = \frac{R_3}{R_1 + R_2} \cdot \frac{V_1}{R_3 + \frac{R_2 R_1}{R_2 + R_1}} = \frac{R_3 V_1}{R_3(R_2 + R_1) + R_2 R_1}$$

MED SIFFROR

$$A = \frac{10 \cdot 1.2 \cdot 10^3}{1.2 \cdot 10^3 (1.2 \cdot 10^3 + 300 \cdot 10^3) + 300 \cdot 10^3 \cdot 1.2 \cdot 10^3} = 8.3 \text{ mA}$$

MEN DET VAR N  $V_c$  VI  $V_{INTE}$  HA, VI VET ATT

$$V_c(t) = V_c(0) + \frac{1}{C} \int_0^t i_3 dt$$

$$\Rightarrow V_c(t) = 0 + \frac{A}{C} \int_0^t e^{-s/\tau} ds = \frac{A}{C} \left[ -\frac{1}{\tau} (e^{-t/\tau} - 1) \right] = \frac{A}{C\tau} (1 - e^{-t/\tau})$$

$$\text{SVAR: } \begin{cases} i_3(t) = 8.3 \cdot 10^{-3} e^{-t/\tau} \text{ A} \\ \tau = 0.24 \text{ sek} \end{cases}$$